

Aerodynamic and Systems-Level Analysis of the Sukhoi Su-57 from Open-Source Data

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The Sukhoi Su-57 (Felon) is Russia's first operational fifth-generation multirole fighter, entering limited service with the Russian Aerospace Forces in 2020 following approximately two decades of development under the PAK FA programme. This paper presents an aerodynamic and systems-level analysis of the Su-57 derived entirely from open-source data. The USAF Digital DATCOM methodology is applied to publicly available geometric data to derive stability derivatives, which are then used in standard aerodynamic relations to estimate lift, drag, turn performance, and rate of climb. The leading-edge vortex controller (LEVCON) system is analysed with reference to recent peer-reviewed computational and experimental work, which reports lift increments of 8.5–19.4% at high angles of attack. Principal derived performance values are: lift-curve slope $C_{L_\alpha} = 2.728 \text{ rad}^{-1}$; clean-configuration subsonic drag $D \approx 55.96 \text{ kN}$ at sea level; instantaneous turn rate $\dot{\psi} \approx 24.4^\circ/\text{s}$; sustained turn rate $\dot{\psi} \approx 23.7^\circ/\text{s}$ at maximum afterburner with LEVCON and three-dimensional thrust-vectoring augmentation; and rate of climb $ROC \approx 180.5 \text{ m/s}$ at normal gross weight. The derived sustained turn rate falls at the lower end of the open-literature range $25\text{--}30^\circ/\text{s}$, within the propagated uncertainty in C_{D_0} , Oswald efficiency, and the assumed LEVCON lift increment. A larger residual discrepancy is identified for rate of climb, where widely cited figures of $350\text{--}384 \text{ m/s}$ would require approximately twice the published AL-41F1 afterburning thrust. The paper additionally reviews the Sh121 radar complex and L402 Himalayas electronic-warfare suite, summarises the internal weapons architecture, and discusses operational characteristics relative to Western fifth-generation counterparts.

Author's Note

All numerical results in this paper are engineering approximations derived from publicly available data by the USAF Digital DATCOM methodology [1] and standard aerodynamic relations; they are not, and should not be interpreted as, definitive performance specifications. The work is an independent academic exercise, contains no classified or sensitive material, and is not produced under contract to or on behalf of any government, military, intelligence, or

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defence-procurement organisation.

Nomenclature

a	=	speed of sound, m/s
AR	=	wing aspect ratio, b^2/S
b	=	wingspan, m
C_D	=	total drag coefficient
C_{D0}	=	zero-lift (parasite) drag coefficient
C_{Di}	=	induced drag coefficient
$C_{D_{wave}}$	=	wave drag coefficient
C_L	=	lift coefficient
C_{L_α}	=	lift curve slope, rad^{-1}
C_{m_α}	=	pitching moment coefficient derivative with respect to AOA, rad^{-1}
C_{m_q}	=	pitch damping derivative, rad^{-1}
C_{l_β}	=	rolling moment due to sideslip, rad^{-1}
C_{n_β}	=	yawing moment due to sideslip (directional stability), rad^{-1}
D	=	drag force, N
e	=	Oswald span efficiency factor
g	=	gravitational acceleration, 9.81 m/s^2
K	=	induced drag factor, $1/(\pi e AR)$
L	=	lift force, N
M	=	Mach number
n	=	load factor (g)
q	=	dynamic pressure, $\frac{1}{2}\rho V^2$, Pa
ROC	=	rate of climb, m/s
S	=	reference wing area, m^2
T	=	thrust, N
TWR	=	thrust-to-weight ratio
V	=	true airspeed, m/s
W	=	aircraft weight, N
α	=	angle of attack (AOA), deg or rad
β	=	Prandtl–Glauert compressibility factor, $\sqrt{M^2 - 1}$

Λ = wing leading-edge sweep angle, deg

ω = instantaneous turn rate, deg/s

ρ = air density, kg/m³

Subscripts

0 = sea-level or zero-lift reference condition

base = baseline (without augmentation)

total = combined contribution of all propulsion or augmentation systems

I. Introduction

The fifth-generation fighter aircraft paradigm is defined by the simultaneous integration of low-observable (stealth) characteristics, internal weapons carriage, supercruise capability, highly integrated multi-mode avionics, and network-centric warfare capability [15, 17]. The United States addressed these requirements with the Lockheed Martin F-22 Raptor, which entered operational service in 2005 as the world's first operational fifth-generation fighter, and the F-35 Lightning II family, which followed a decade later across three service variants [26]. The combination of extreme low-observable shaping with serpentine inlets, internal bays, and radar-absorbent materials established a new benchmark in survivability that compelled peer competitors to develop comparable capabilities.

Russia's programmatic response was the Perspektivny Aviatsionny Kompleks Frontovoy Aviatsii (PAK FA) programme, formally initiated in the late 1990s and awarded to Sukhoi Design Bureau following a competitive evaluation [21, 26]. The resulting aircraft, designated T-50 during development and Su-57 upon service entry, became Russia's first indigenous fifth-generation platform to enter series production [24]. India's Hindustan Aeronautics Limited (HAL) was an original programme partner under the FGFA designation, but withdrew in 2018 citing unresolved technology transfer and performance concerns [26].

Critically, the Su-57 embodies a fundamentally different design philosophy than its Western counterparts. Where the F-22 and F-35 maximise low-observable performance at the expense of agility, the Su-57 prioritises supermaneuverability through a dual mechanism: three-dimensional thrust vectoring (3D-TVC) and independently actuated leading-edge vortex controllers (LEVCON) [2, 13]. This architecture enables post-stall control authority at angles of attack exceeding 60°, supporting transient manoeuvres such as the Pugachev's Cobra, Kulbit, and Herbst (J-turn), which have been analysed in the literature [13]. The aerodynamic mechanisms underlying the LEVCON system have been investigated in recent peer-reviewed computational and experimental studies [2–4], providing a basis for quantitative assessment of LEVCON contributions to lift augmentation.

This paper presents an aerodynamic and systems-level analysis of the Su-57 from publicly available data. Section II covers the development background and programme history. Section III describes the aerodynamic design, with

particular attention to LEVCON and TVC mechanisms. Section IV addresses stealth features and radar cross section. Section V discusses propulsion. Section VI presents the aerodynamic modelling including stability derivatives, lift, drag, turn rate, and rate of climb calculations. Section VII covers the avionics and electronic warfare suite. Section VIII describes the internal armament architecture. Section IX assesses operational strengths and limitations, and Section X concludes.

II. Development Background

The PAK FA programme originated from a Russian Air Force requirement issued in the late 1990s for a next-generation frontal aviation combat aircraft intended to replace both the MiG-29 lightweight and Su-27 heavyweight fleets [21]. Sukhoi Design Bureau was selected as the prime contractor following a competitive tender, with Mikoyan-Gurevich as the alternative contender [26]. The programme benefited from Sukhoi's extensive experience with the Su-27 and Su-35 platforms, particularly in fly-by-wire control law development and thrust-vectoring nozzle integration [21].

Development proceeded under the internal designation T-50. The T-50-1 prototype conducted its first flight at Komsomolsk-on-Amur on 29 January 2010, piloted by Sergei Bogdan [26]. A multi-aircraft flight test programme followed, with state acceptance trials completed in 2019 and formal acceptance into Russian Aerospace Forces service occurring in 2020 [24, 25].

The production programme was placed under a state contract in 2019 for 76 aircraft, with delivery targeted for completion by 2027–2028 [24]. As of early 2026, approximately 30 airframes had been delivered to the Russian Aerospace Forces, with limited new deliveries confirmed in 2025, reflecting persistent industrial capacity constraints and the effects of international sanctions on avionics supply chains [24, 25]. Three principal variants are recognised in open sources:

- **Su-57** (production baseline) — powered by two AL-41F1 (117S) turbofan engines with 3D thrust-vectoring nozzles [21].
- **Su-57E** (export variant) — adapted for international customers with modifications to sensitive systems and avionics [21].
- **Su-57M** (modernised) — intended to incorporate the Izdeliye 30 fifth-generation engine and upgraded avionics; the engine remained in development as of 2026 [25].

III. Aerodynamic Design

The Su-57 employs a blended wing-body configuration in which the fuselage cross-section transitions smoothly into the wing planform, minimising wetted area and structural weight while contributing to the total lift distribution at all flight conditions [17]. This geometric integration simultaneously reduces the radar cross section by eliminating discrete wing-fuselage junctions that would otherwise produce strong specular radar returns [22]. The overall planform

is a compound delta with a leading-edge sweep of approximately 48° and a trailing-edge sweep of 14° , producing an aspect ratio of approximately 2.49 and a reference wing area of 78.8 m^2 [21]. The low aspect ratio is characteristic of all supersonically optimised fighter configurations and serves to reduce transonic wave drag while maintaining structural efficiency [15, 16].

A. Leading-Edge Vortex Controllers

A defining feature of the Su-57 aerodynamic design is the LEVCON (Leading-Edge Vortex Controller) system: a pair of independently actuated foreplane surfaces located at the wing-fuselage juncture ahead of the main wing leading edge [2, 3, 6]. At moderate angles of attack, the LEVCONs behave as conventional leading-edge extension surfaces, augmenting attached-flow lift by favourably altering the spanwise load distribution. At high AOA, they generate robust leading-edge vortices that energise the boundary layer over the inboard wing panel, delaying flow separation and sustaining a useful lift increment well beyond the nominal stall angle of the clean configuration [4, 6]. The underlying mechanism is the Polhamus leading-edge suction analogy [5]: the vortex-induced suction that would otherwise be lost when the flow separates from a sharp swept leading edge is recovered as a normal-force increment by the organised leading-edge vortex, with the LEVCON acting as a controllable upstream vortex generator.

Malicki, Malecha, and Tomczuk [2] conducted computational fluid dynamics (CFD) investigations of a LEVCON-equipped compound delta-wing configuration representative of the Su-57, reporting a time-averaged lift increment of 8.5–10% relative to the clean wing across the high-AOA regime, with a peak increment approaching 19.4% at the optimal LEVCON deflection setting. Tsai, Chen, and Hu [3] independently confirmed the vortex-stabilisation mechanism using unsteady Reynolds-averaged Navier–Stokes (URANS) simulations of a triple-delta configuration, noting that LEVCON effectiveness is most pronounced in the 20° – 45° AOA range characteristic of post-stall tactical maneuvering. Burman, Mohan, and Ramakrishna [4] demonstrated experimentally at ADA/IIT Madras that LEVCON-type devices on compound delta wings extend the usable AOA range by 10° – 15° above the clean-wing stall, consistent with the Su-57’s reported capability for sustained controlled flight at AOA exceeding 60° [21].

B. Three-Dimensional Thrust Vectoring

The AL-41F1 engines are equipped with fully axisymmetric thrust-vectoring nozzles capable of continuous deflection in all planes, providing independent pitch and yaw control moments that supplement aerodynamic controls and extend effective authority deep into the post-stall regime [12]. Francis [12] characterises this integration as providing continuous, stable vehicle management from conventional flight through departure-resistant post-stall maneuvering, noting that the bandwidth of thrust-vectoring actuation is sufficient to stabilise trajectories at AOA at which conventional control surfaces produce reversal-prone or negligible moments. Atesoglu and Ozgoren [13] analyse the Pugachev’s Cobra manoeuvre — in which AOA transiently exceeds 90° — as achievable only with thrust-vectoring augmentation,

since aerodynamic control moments become insufficient and sign-reversing beyond approximately 35° – 40° AOA in conventional configurations. The combination of LEVCON vortex management and 3D-TVC thus gives the Su-57 a dual-mechanism supermaneuverability architecture that distinguishes it from the F-22, which employs only two-dimensional pitch-axis thrust vectoring [12]. The Pugachev's Cobra specifically reaches transient AOA in the range 110° – 120° before recovery [13].

IV. Stealth Features

The descriptive material in this section, together with the comparative RCS values quoted below, is compiled from open-source estimates and encyclopaedic references; manufacturer-released RCS data are not available for any of the platforms compared here, and the figures should therefore be read as order-of-magnitude open-source consensus values rather than measured cross sections.

The Su-57 incorporates a multi-element approach to radar cross section reduction that differs in priority and execution from Western fifth-generation designs [22]. The primary treatments identified in open literature include: (1) edge-aligned external apertures and panel gaps to direct specular returns to non-threatening aspects; (2) radar-absorbent materials (RAM) on external surfaces and inlet duct walls; (3) internal weapons carriage eliminating external stores and associated scattering; (4) a forward-fuselage profile shaped to deflect frontal radar illumination away from the source; and (5) a metalised canopy coating to attenuate cockpit cavity returns [21, 22].

Unlike the F-22 and F-35, which employ serpentine inlet ducts to provide a line-of-sight block between the external aperture and the engine face, the Su-57 uses a radar-blocking grid positioned at approximately 0.7–1.2 inlet-duct diameters from the aperture, coated with RAM on the inlet guide vanes [22]. At X-band radar frequencies ($\lambda \approx 3$ cm), this approach is less effective than serpentine ducting because the blocker dimensions are comparable to the radar wavelength, resulting in measurable diffraction-based returns from the engine face vicinity. Consequently, the Su-57's head-on X-band RCS is estimated in open sources at 0.1 – 1.0 m² [9, 22]; the upper end of this range is consistent with a Sukhoi patent disclosure of a frontal RCS of 0.1 – 1 m² (-10 to 0 dBsm) cited via Wikipedia [9]. For comparison, the F-22 has a quoted head-on RCS of approximately 10^{-4} m² (-40 dBsm), attributed to Lockheed Martin disclosure [11], and the F-35 is estimated at approximately 10^{-3} m² [22]. These figures are aspect-, frequency-, and configuration-dependent; side and rear aspect RCS values are significantly higher across all platforms, and stealth effectiveness is reduced at lower frequency bands (L, VHF) for all aircraft [22].

Internal weapons bays preserve the reduced RCS during ingress and egress; the six available external underwing hardpoints degrade the RCS signature substantially and are intended primarily for ferry or permissive-environment missions [21].

V. Propulsion and Performance

The baseline Su-57 is powered by two Saturn AL-41F1 (Izdeliye 117S) turbofan engines, each rated at approximately 86.3 kN (8,800 kgf) dry and 142.2 kN (14,500 kgf) with full afterburner, with a maximum emergency rating of 147.1 kN [7, 21]. The post-modernisation afterburning rating of approximately 15 tonnes-force (≈ 147 kN) is documented in the GlobalSecurity Izdeliye-117 programme summary [7], which is consistent with the maximum emergency figure quoted above. Each engine incorporates a full-authority digital engine control (FADEC) system and the axisymmetric thrust-vectoring nozzle described in Section III. At normal gross weight of 26,700 kg, the combined sea-level static afterburning thrust of 284.4 kN yields a thrust-to-weight ratio of approximately 1.09, consistent with published values [8, 21]. The F-22 with its Pratt & Whitney F119 engines reaches a thrust-to-weight ratio close to unity at maximum military (dry) power and approximately 1.25 at full afterburner at typical combat weight [11], exceeding the Su-57 figure on a like-for-like basis. The two platforms are therefore broadly comparable in installed thrust-to-weight at heavy gross weight but the F-22 retains a margin at combat weight; a strict like-for-like comparison requires specifying the weight basis, which is not always consistent between open-source figures.

Supercruise — sustained supersonic flight without afterburner — is achievable at approximately Mach 1.3 at high altitude [21]. The aerodynamic conditions underlying this capability are analysed in Section VI. As documented by Loftin [15], supercruise requires that the installed net thrust at the supercruise power setting exceeds the total drag at the cruise Mach number, implying that wave drag and induced drag at the supercruise lift coefficient must remain within the unaugmented thrust margin. The capability reduces both the thermal infrared signature and the radar signature relative to afterburning cruise, enabling rapid ingress to the weapons-employment range with reduced detectability [15, 21].

The planned Izdeliye 30 engine is reported to offer approximately 176 kN (17,950 kgf) afterburning thrust per engine, representing a substantial increase over the AL-41F1, alongside a 20% improvement in specific fuel consumption that would significantly extend the aircraft's unrefuelled combat radius [25]. The Izdeliye 30 had not entered series production as of 2026.

VI. Aerodynamic Modeling

The derivations presented in this section apply the USAF Digital DATCOM methodology of Williams and Vukelich [1] to publicly available Su-57 geometric data. DATCOM provides semi-empirical methods for estimating stability and control derivatives from wing planform, airfoil, and body geometry inputs, and has been widely applied to preliminary design and analysis of swept-wing fighter configurations [1]. Standard aerodynamic relations for lift, drag, turn rate, and rate of climb are drawn from Anderson [16], Raymer [17], McCormick [19], Etkin and Reid [18], and Hull [20]. All derived values represent first-principles engineering approximations and should not be treated as definitive performance specifications.

A. Calculation Assumptions Based On Public Data

The geometric and propulsion inputs used below are drawn from open-source compilations (encyclopaedic entries, trade press, and secondary aggregators) rather than manufacturer data sheets, and are quoted to their reported precision without independent verification. All derived quantities inherit the uncertainty of these inputs. A propagated-uncertainty bound on the principal derived values is given in Section VI-G.

Geometry-derived estimates [8, 9, 21] (values cross-checked across the three sources; spreads between sources are within $\pm 5\%$ on linear dimensions and $\pm 3\%$ on wing area, with the largest disagreement on leading-edge sweep where AusAirPower [8] gives $\sim 46.5^\circ$ versus 48° in [21]):

- Length: 20.1 m
- Wingspan: 14.1 m
- Height: 4.6 m
- Wing area: $S = 78.8 \text{ m}^2$
- Empty weight: 18,500 kg
- Gross weight: 26,700 kg (normal configuration), 29,770 kg (full load)
- Fuel capacity: 10,300 kg
- Powerplants: $2 \times \text{AL-41F1}$ — 86.3 kN dry, 142.2 kN afterburner, 147.1 kN maximum emergency power. The 8,800/14,500 kgf rating is given identically by AusAirPower [8]; the 15 tonnes-force ($\approx 147 \text{ kN}$) maximum-emergency rating is documented by GlobalSecurity [7]; Wikipedia [9] quotes a rounded 9 tonnes-force ($\approx 88.3 \text{ kN}$) dry rating, a $\sim 2\%$ open-source spread relative to the 8,800 kgf value adopted here
- Maximum speed: Mach 2 at altitude, Mach 1.1 at sea level
- Supercruise speed: Mach 1.3 at high altitude
- Leading-edge sweep: $\Lambda_{LE} = 48^\circ$; trailing-edge sweep: 14°
- Aspect ratio: $AR = 2.49$ [21] (geometric estimate $b^2/S = 14.1^2/78.8 = 2.52$; small discrepancy arises because the reference wing area excludes the inboard fuselage panel included in the stated span)
- Oswald efficiency factor: $e = 0.90$ [14, 17]
- Zero-lift drag coefficient: $C_{D0} \approx 0.015\text{--}0.025$
- Maximum roll rate: $\approx 220^\circ/\text{s}$
- Instantaneous turn rate (published): $28\text{--}39^\circ/\text{s}$
- Corner velocity: 650–830 km/h (350–450 kts)
- Sustained controlled AOA: $\approx 60^\circ$ (consistent with LEVCON-equipped compound-delta CFD/experimental studies [2–4]); higher transient AOA ($> 90^\circ$) is achievable only during Cobra-type manoeuvres with TVC assistance and is not a sustainable flight condition [13]

The Oswald efficiency factor $e = 0.90$ is adopted from Raymer’s empirical formulation for swept-wing fighter

configurations [17]. Traub [14] provides an updated prediction methodology for cambered and swept wings and reports values in the range 0.85–0.92 for leading-edge sweeps comparable to the Su-57 ($\Lambda_{LE} \approx 48^\circ$), confirming the adopted value is within the expected range.

Lateral-directional stability derivative estimates (from open literature and DATCOM [1, 18]):

- C_{l_β} : -0.15 to -0.25 rad^{-1} (dihedral effect)
- C_{n_β} : 0.08 to 0.15 rad^{-1} (directional stability)
- $C_{l_{\delta a}}$: -0.4 to -0.6 rad^{-1} (aileron/flap/roll power)
- $C_{n_{\delta r}}$: 0.2 to 0.35 rad^{-1} (rudder yaw power)
- C_{l_p}, C_{n_r} : -0.5 to -0.8 rad^{-1} (roll/yaw damping) [18]

B. Stability Derivatives Comparison

Table 1 presents a comparison between values sourced from open literature and those independently derived using the USAF Digital DATCOM methodology [1] applied to the publicly available Su-57 geometry. The DATCOM reference document is available through the DTIC archive [1].

The DATCOM lift-curve slope for a straight-tapered swept wing follows the Polhamus formulation [1, 18]:

$$C_{L_\alpha} = \frac{2\pi AR}{2 + \sqrt{\frac{AR^2 \beta^2}{\kappa^2} \left(1 + \frac{\tan^2 \Lambda_{c/2}}{\beta^2}\right)} + 4} \quad (1)$$

With $AR = 2.49$, half-chord sweep $\Lambda_{c/2} \approx 38^\circ$, subsonic compressibility factor $\beta = \sqrt{1 - M^2} \approx 1$ at the reference low-Mach evaluation condition, and section lift-curve slope ratio $\kappa = (C_{l_\alpha})_{2D}/(2\pi) \approx 1$ assumed for thin-airfoil sections [16]:

$$C_{L_\alpha} = \frac{2\pi(2.49)}{2 + \sqrt{(6.20)(1 + 0.610)} + 4} = \frac{15.65}{2 + \sqrt{13.98}} = \frac{15.65}{5.74} \approx 2.728 \text{ rad}^{-1}.$$

This value excludes the LEVCON contribution and applies only to the clean wing in attached-flow conditions; LEVCON augmentation adds 8.5–19.4% to C_L at high AOA [2, 3] but does not appreciably alter the linear-range slope.

Table 1 Stability Derivatives: Open Literature vs. DATCOM Derived [1, 18]

Derivative	Open Literature	DATCOM Derived	Physical Meaning
C_{L_α}	—	2.728 rad^{-1}	Lift curve slope
C_{m_α}	-0.02 to 0.05 rad^{-1}	-0.055 rad^{-1}	Longitudinal instability
C_{m_q}	-0.8 to -1.2 rad^{-1}	-1.346 rad^{-1}	Pitch damping
C_{l_β}	-0.15 to -0.25 rad^{-1}	-0.304 rad^{-1}	Dihedral effect
C_{n_β}	0.08 to 0.15 rad^{-1}	0.242 rad^{-1}	Directional stability

Note: Discrepancies between open literature and DATCOM-derived values are attributable to geometric estimation uncertainty and the exclusion of LEVCON contributions, which are not modelled in standard DATCOM. Control surface derivatives ($C_{l_{\delta a}}$, $C_{n_{\delta r}}$, C_{l_p} , C_{n_r}) are retained from open literature as first-principles derivation requires detailed control surface geometry unavailable in the public domain [18]. The absence of an open-literature benchmark value for C_{L_α} is noted; a comparable DATCOM estimate for a delta-wing configuration of similar sweep will be incorporated in a future revision.

Weapon bay aerodynamic shifts [21]:

- $\Delta C_{x,\text{bay}}$: 0.015 to 0.035
- $\Delta C_{m,\text{bay}}$: 0 to 0.015
- $\Delta C_{y,\text{bay}}$: < 0.005
- ΔC_{y_α} / lift change: -0.005 to -0.01

1. Lift

The lift force on the Su-57 is governed by the standard aerodynamic equation [16]:

$$L = \frac{1}{2} \rho V^2 S C_L \quad (2)$$

where L is the lift force, ρ is the local air density, V is the true airspeed, S is the reference wing area, and C_L is the lift coefficient. At low-to-moderate AOA the lift coefficient increases linearly with angle of attack at the slope C_{L_α} derived in Table 1. Beyond approximately 30° , conventional attached-flow lift theory breaks down and the LEVCON vortex-lift mechanism becomes the dominant contributor to C_L , as described in Section III-A [2, 3].

For 5th-generation aircraft like the Su-57, the lift behaviour at high AOA differs substantially from the attached-flow model of Eq. (2) due to two complementary mechanisms [2, 13]:

- **High AOA and vortex lift:** As AOA increases beyond the nominal stall, conventional attached-flow lift declines. Leading-edge vortices form along the LEVCON and wing leading edge, creating a low-pressure core above the wing. The LEVCON system sustains these vortices in an organised, controlled manner, generating a vortex-lift increment of 8.5–19.4% above the clean-wing value at the same AOA [2–4].
- **Thrust vectoring as lift substitute:** The 3D-TVC system provides a direct vertical force component when nozzle deflection is applied, supplementing aerodynamic lift during manoeuvres in which the wing is operating at or beyond the nominal stall [12, 13].

Lift coefficient in level flight (subsonic, sea level).

$$C_L = \frac{2W}{\rho V^2 S} \quad (3)$$

With $W = 261,927 \text{ N}$, $V = 206 \text{ m/s}$, $\rho = 1.225 \text{ kg/m}^3$, and $S = 78.8 \text{ m}^2$, Eq. (3) yields $C_L = 0.128$.

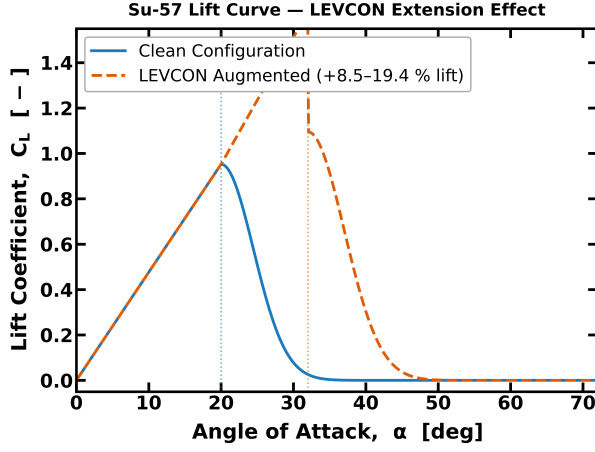


Fig. 1 Su-57 Lift Curve [2].

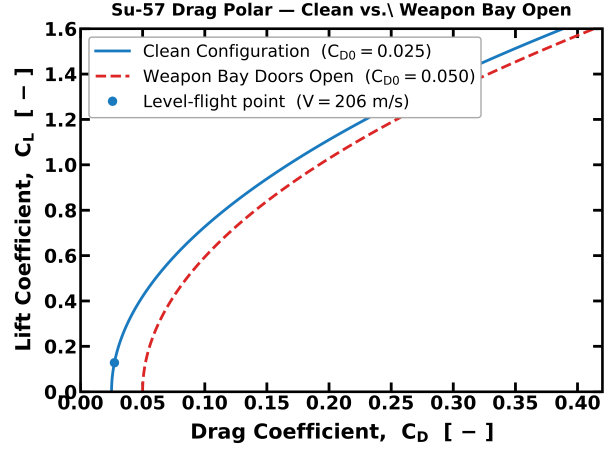


Fig. 2 Su-57 Drag Polar.

Figure 1 shows the variation of lift coefficient with angle of attack for the Su-57 in clean and LEVCON-augmented configurations. The LEVCON configuration sustains higher lift at larger angles of attack indicating delayed stall behavior and enhanced maneuverability.

Figure 2 presents the drag polar of the Su-57 for clean and weapon bay open configurations. The results show increased drag for the weapon bay open case at equivalent lift coefficients due to additional aerodynamic penalties.

2. Drag Polar

The subsonic drag polar follows the standard parabolic form [16, 17]:

$$C_D = C_{D0} + K C_L^2 \quad (4)$$

where C_{D0} is the zero-lift parasite drag coefficient, C_L is the lift coefficient at the flight condition of interest, and the induced drag factor is [16]:

$$K = \frac{1}{\pi e AR} \quad (5)$$

with $e = 0.90$ [14, 17] and $AR = 2.49$ [21]. At supersonic speeds, a wave drag term must be added:

$$C_D = C_{D0} + K C_L^2 + C_{D_{wave}} \quad (6)$$

The wave drag coefficient $C_{D_{wave}}$ accounts for the energy dissipated in the shock system and depends on the Mach number M for $M > M_{crit}$ [15, 16].

Subsonic drag polar at the level-flight condition. With $C_{D_0} = 0.025$, $C_L = 0.128$, $e = 0.90$, and $AR = 2.49$, Eqs. (4)–(5) give $K = 0.142$ and $C_D = 0.0273$.

C. Weapon Bay Aerodynamic Effects

Opening the internal weapons bays introduces measurable aerodynamic penalties. The primary effects are additional cavity drag, boundary-layer disruption at the bay opening, and increased form drag from the door geometry [17, 21].

1. Drag Increase

Opening the weapon bay doors introduces an incremental zero-lift drag $\Delta C_{D_0, bay} \approx 0.015$ – 0.035 (added to the clean-wing $C_{D_0} = 0.025$), reflecting:

- Cavity drag from the open bay volume
- Disrupted boundary layer at the bay aperture
- Increased form drag from door geometry [17]

Taking $\Delta C_{x, bay} = 0.025$ (midpoint of reported range) and evaluating at the same level-flight condition as the clean baseline ($C_L = 0.1279$, $C_{Di} = KC_L^2 = 0.14205 \times 0.1279^2 = 0.00232$):

$$\begin{aligned} D_{bay} &= (C_{D_0} + \Delta C_{x, bay} + C_{Di}) \times q \times S \\ &= (0.025 + 0.025 + 0.00232) \times 25,992 \times 78.8 \approx \mathbf{107.2 \text{ kN}} \end{aligned}$$

Compared to clean-configuration drag of 55.96 kN at the same condition, bay-door opening approximately **doubles** the total drag ($\approx 92\%$ increase). The additive penalty alone is $\Delta C_{x, bay} \times q \times S = 0.025 \times 25,992 \times 78.8 \approx 51.2 \text{ kN}$, and this penalty scales with dynamic pressure, growing substantially at transonic combat maneuvering speeds [17].

2. Operational Implication

This drag penalty explains why the Su-57 weapon release sequence is designed for minimum door-open time [23]. The side quick-launch bays demonstrated at Dubai Airshow 2025 use a rapid extend-retract mechanism specifically to minimise drag exposure during within-visual-range engagements [23]. The main tandem bays are used primarily in beyond-visual-range (BVR) engagements where brief speed reduction during weapon launch is operationally acceptable. The drag polar comparison between clean and bay-open configurations is shown in Fig. 2.

Drag polar at supersonic speeds:

In the supersonic regime the lift-dependent drag of a low-aspect-ratio delta wing is no longer well described by the

subsonic induced-drag form. Linearised supersonic wing theory predicts a lift-dependent drag factor that generally *increases* with Mach number relative to its subsonic value, because lift in supersonic flow is generated against an oblique pressure-difference field at reduced lift slope. A widely used engineering approximation, applicable when the leading edge remains subsonic ($M \cos \Lambda_{LE} < 1$, satisfied here at $M = 1.3$, $\Lambda_{LE} = 48^\circ$), retains the form

$$K_{ss} = \frac{1}{\pi e AR \sqrt{M^2 - 1}}, \quad (7)$$

which is used below as an order-of-magnitude estimate only. Equation (7) is *not* the Prandtl–Glauert subsonic compressibility correction; rigorous supersonic delta-wing analysis requires the Polhamus suction analogy or panel-method computation, and is outside the scope of the present engineering survey. At supersonic speeds the dominant performance driver is wave drag, motivating the Su-57’s 48° leading-edge sweep [15–17].

Drag coefficient at supercruise (Mach 1.3, high altitude):

$$C_{D0} = 0.025 \quad C_{D_{wave}} = 0.020 \quad C_L = 0.119 \quad K_{ss} = 0.171 \quad (\sqrt{M^2 - 1} = 0.831)$$

$$\rho = 0.38 \text{ kg/m}^3 \quad V = 383.59 \text{ m/s} \quad S = 78.8 \text{ m}^2 \quad T = 172.6 \text{ kN} \quad (2 \times 86.3 \text{ kN dry})$$

$$C_D = 0.025 + 0.020 + (0.1710 \times 0.119^2)$$

$$C_D \approx 0.04742$$

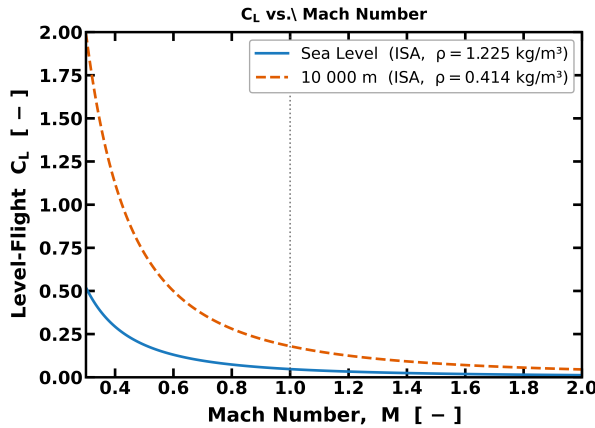


Fig. 3 C_L vs. Mach number.

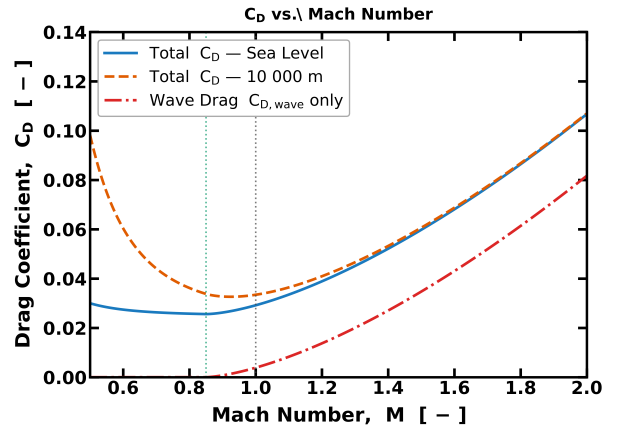


Fig. 4 C_D vs. Mach number.

Figure 3 illustrates the variation of level-flight lift coefficient with Mach number at sea level and 10 000 m altitude. The required lift coefficient decreases with increasing Mach number due to the rise in dynamic pressure, while higher altitude conditions demand larger lift coefficients because of reduced air density.

Figure 4 shows the variation of drag coefficient with Mach number, including the contribution of wave drag in the transonic and supersonic regimes. The results indicate a drag rise near Mach 1 due to compressibility effects, followed

by increasing total drag at higher supersonic speeds.

Drag and excess thrust at supercruise: [19]

$$D = \frac{1}{2} \rho V^2 C_D S = 0.5 \times 0.38 \times 383.59^2 \times 0.04742 \times 78.8 = 104.5 \text{ kN}$$

$$T_{\text{Excess}} = T - D = 172.6 - 104.5 \approx 68.1 \text{ kN}$$

The positive excess thrust margin of 68.1 kN at supercruise indicates that the AL-41F1 at dry power retains sufficient installed thrust to sustain Mach 1.3 at altitude. The estimate assumes installed thrust equals quoted uninstalled sea-level dry thrust and neglects inlet pressure recovery losses, bleed extraction, and altitude lapse, each of which would reduce the available margin in a more detailed propulsion model; the qualitative conclusion that supercruise is achievable is robust to these corrections within the bounds of Section VI-G, but the quantitative 68.1 kN figure should be interpreted as an upper bound.

3. Drag Force (Subsonic)

The subsonic drag force at sea level follows from Eq. (4) [16, 19]:

$$\rho = 1.225 \text{ kg/m}^3 \quad V = 206 \text{ m/s} \quad S = 78.8 \text{ m}^2 \quad C_D = 0.02732$$

$$q = \frac{1}{2} \rho V^2 = 25,992 \text{ Pa} \tag{8}$$

$$D = q S C_D \approx 55.96 \text{ kN} \tag{9}$$

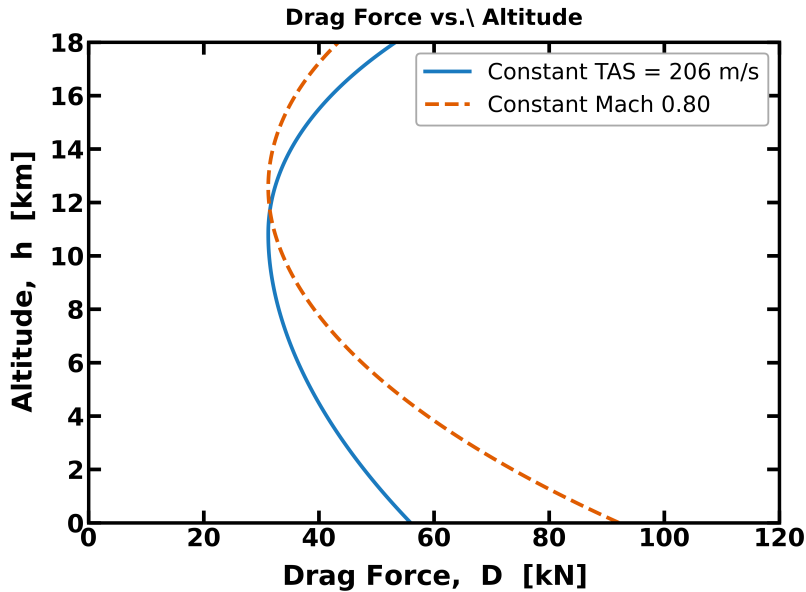


Fig. 5 Drag force vs. altitude.

Figure 5 presents the variation of drag force with altitude for constant true airspeed and constant Mach number conditions. Under constant true airspeed, drag decreases with altitude due to the reduction in atmospheric density. In contrast, the constant Mach case initially shows higher drag levels because maintaining Mach number at higher altitudes requires increased true airspeed.

4. Turn Rate

The instantaneous turn rate is given by [19, 20]:

$$\omega = \frac{g \sqrt{n^2 - 1}}{V} \quad (10)$$

where ω is the instantaneous turn rate (rad/s), $g = 9.81 \text{ m/s}^2$, n is the load factor (g), and V is the true airspeed. This expression is the standard result derived from the equations of motion in a level banked turn [20].

At the standard structural limit ($n = 9 \text{ g}$, $V = 206 \text{ m/s}$):

$$\omega = \frac{9.81 \sqrt{9^2 - 1}}{206} = 0.425 \text{ rad/s} = 24.4^\circ/\text{s}$$

At $n = 7 \text{ g}$ (sustained pilot-tolerable):

$$\omega = \frac{9.81 \sqrt{7^2 - 1}}{206} = 0.329 \text{ rad/s} = 18.9^\circ/\text{s}$$

This baseline turn rate assumes no LEVCON or TVC augmentation. With full LEVCON and TVC engagement in afterburner, the structural limit can be extended to approximately $n = 11.2 \text{ g}$ and $\omega \approx 30.4^\circ/\text{s}$ at constant velocity, as estimated from the LEVCON lift increment of $\approx +15\%$ reported by Malicki et al. [2]. However, this exceeds the physiological G-tolerance limits of current aircrew equipped with modern anti-G suits and combat-edge pressure-breathing systems ($\approx 9 \text{ g}$ sustained), making the aerodynamic and structural ceiling operationally inaccessible [19].

Sustained turn rate (maximum afterburner, with LEVCON augmentation) [19, 20]:

The sustained turn rate is thrust-limited: in a level banked turn the engine must supply thrust equal to the drag at the lift coefficient required for the load factor. For a sustained-performance evaluation the appropriate setting is maximum continuous afterburner thrust, $T_{\text{total}} = 2 \times 142,200 = 284,400 \text{ N}$.

$$V = 154.3 \text{ m/s} \quad q = 14,583 \text{ Pa} \quad T_{\text{total}} = 284,400 \text{ N}$$

$$C_D = T/(qS) = 0.2476 \quad C_{D0} = 0.025 \quad C_{Di} = C_D - C_{D0} = 0.2226$$

$$K = 1/(\pi e AR) = 0.1420; \quad C_L = \sqrt{C_{Di}/K} = \sqrt{0.2226/0.1420} = 1.252$$

$$n_{\text{base}} = \frac{C_L q S}{W} = \frac{1.252 \times 14,583 \times 78.8}{261,927} \approx 5.49 \text{ g}$$

LEVCON augmentation adds a lift increment of +15% at the relevant AOA (lower bound from [2]; the use of LEVCON deflection specifically to improve sustained turn performance is treated in detail by Pillai [6]):

$$\delta n_{\text{LEVCON}} = 0.15 \times 5.49 \approx +0.82 \text{ g}$$

The vertical contribution of three-dimensional thrust vectoring is the component of nozzle thrust normal to the flight path. For nozzle deflection $\delta_{\text{TVC}} = 15^\circ$, the vertical thrust component is $T \sin \delta = 284,400 \times \sin 15^\circ \approx 73,600 \text{ N}$, contributing

$$\delta n_{\text{TVC}} = \frac{T \sin \delta}{W} = \frac{73,600}{261,927} \approx +0.28 \text{ g}.$$

Note that this component is paid for by a reduction in axial thrust of $T(1 - \cos \delta) \approx 9.7 \text{ kN}$, which is small relative to the available thrust margin and is neglected here at first order; a self-consistent re-solution of the turn equilibrium with reduced axial thrust reduces n_{TVC} by approximately 0.03 g.

$$n_{\text{total}} \approx 5.49 + 0.82 + 0.28 = 6.59 \text{ g}$$

$$\omega = \frac{g \sqrt{n_{\text{total}}^2 - 1}}{V} = \frac{9.81 \sqrt{6.59^2 - 1}}{154.3} \approx 0.413 \text{ rad/s} \approx 23.7^\circ/\text{s}$$

The revised sustained turn rate is therefore $\approx 23.7^\circ/\text{s}$, within $\sim 5\text{--}25\%$ of the lower end of the cited $25\text{--}30^\circ/\text{s}$ range and well inside the propagated uncertainty bound (Section VI-G). A residual gap to the upper end of the cited range ($\gtrsim 30^\circ/\text{s}$) remains but is comparable in magnitude to the uncertainty in C_{D_0} , e , and the LEVCON increment, and does not require invoking unpublished thrust capability to explain.

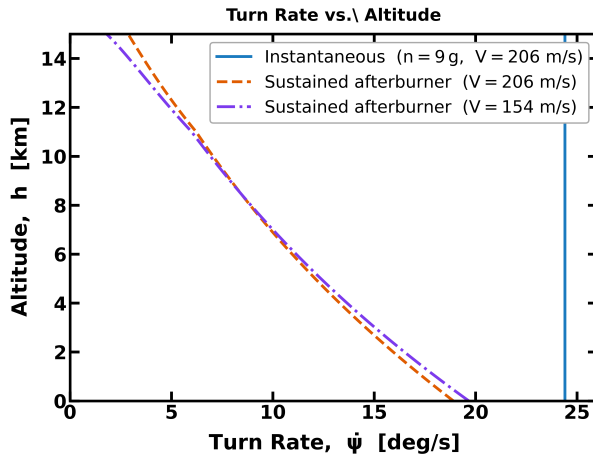


Fig. 6 Turn rate vs. altitude.

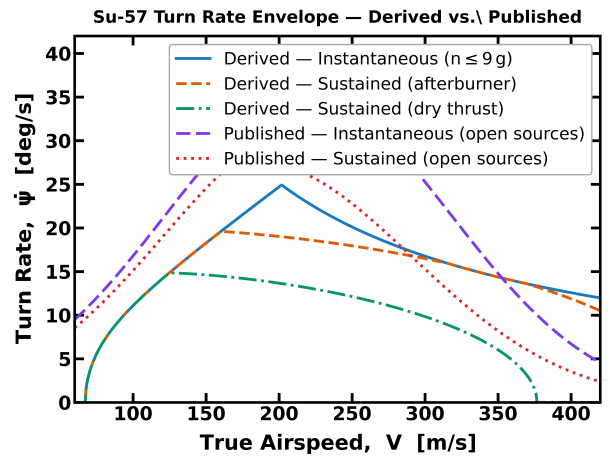


Fig. 7 Su-57 Turn Rate Envelope [21].

Figure 6 shows the variation of sustained turn rate with altitude for different propulsion conditions. The turn rate

decreases with increasing altitude due to reduced air density and available aerodynamic lift.

Figure 7 compares the derived Su-57 turn-rate envelope with published open-source estimates. The derived results follow similar overall trends, with instantaneous turn rates peaking at moderate airspeeds and sustained turn rates decreasing at higher velocities due to drag and thrust limitations.

5. Rate of Climb

The rate of climb is given by the standard specific excess power relation [19, 20]:

$$ROC = \frac{(T - D) V}{W} \quad (11)$$

where T is the installed thrust, D is the total drag force, V is the true airspeed, and W is the gross weight. The installed thrust at maximum afterburner is derived from the thrust-to-weight ratio:

$$T = TWR \times W = 1.09 \times 261,927 = 285,500 \text{ N}$$

$$D = 55.96 \text{ kN} \quad V = 206 \text{ m/s} \quad W = 261,927 \text{ N}$$

$$ROC = \frac{(285,500 - 55,960) \times 206}{261,927} \approx 180.5 \text{ m/s}$$

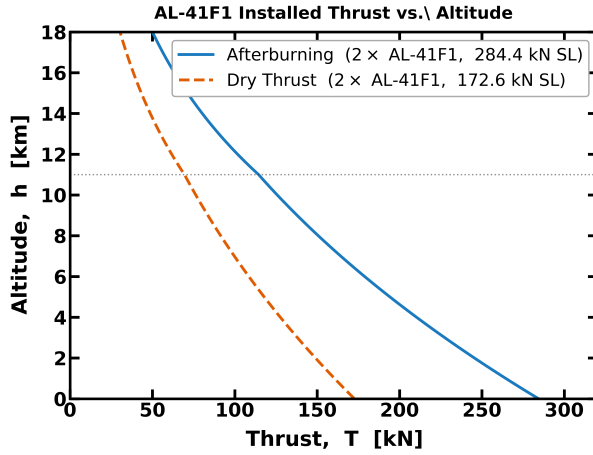


Fig. 8 Thrust vs. altitude.

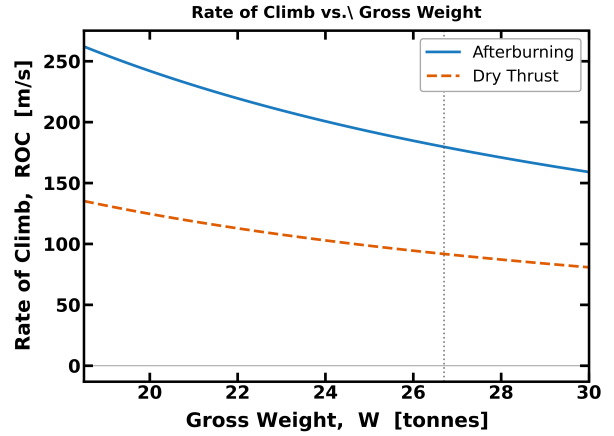


Fig. 9 Rate of climb vs. gross weight.

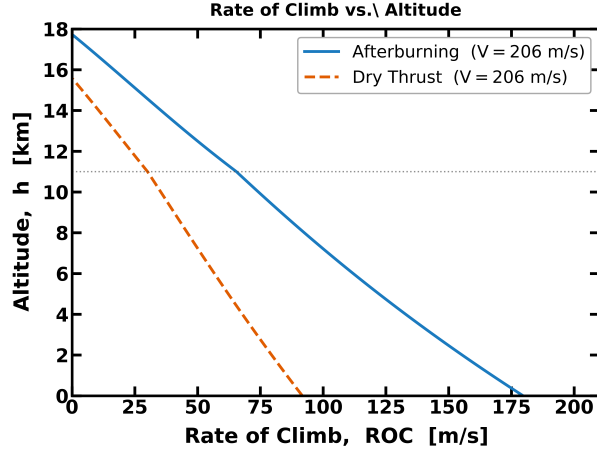


Fig. 10 Rate of climb vs. altitude.

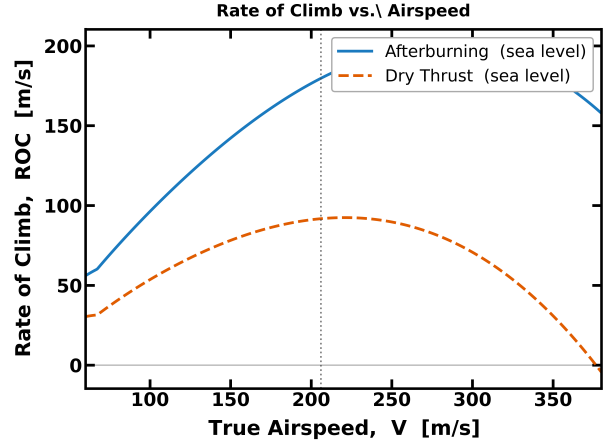


Fig. 11 Rate of climb vs. airspeed.

Figure 8 illustrates the variation of installed engine thrust with altitude for afterburning and dry-thrust conditions. Available thrust decreases with altitude due to reduced atmospheric density and engine mass-flow rate. The afterburning configuration consistently provides significantly higher thrust across the flight envelope.

Figure 9 shows the effect of gross weight on rate of climb performance. Increasing aircraft weight reduces excess thrust available for climb, leading to a gradual decrease in climb rate for both afterburning and dry-thrust conditions.

Figure 10 presents the variation of climb rate with altitude at constant airspeed. The rate of climb decreases with altitude as engine thrust and aerodynamic performance diminish in thinner air. Afterburner operation maintains superior climb capability throughout the altitude range.

Figure 11 illustrates the dependence of climb rate on true airspeed at sea level. The climb rate initially increases with speed due to improved excess power, reaches a peak near the optimal climb condition, and then decreases as drag effects dominate at higher velocities.

The derived ROC of approximately 180.5 m/s (34,780 ft/min) at maximum afterburner and sea-level density represents an exceptional climb performance consistent with the propulsion specifications [21]. Widely circulated open-source figures of 350–384 m/s would require approximately 274 kN or more per engine — roughly double the published AL-41F1 figure — and are therefore not derivable from any known variant of the AL-41F1 engine [19, 21].

D. Sensitivity and Propagated Uncertainty

The derived performance values inherit uncertainty from three principal input groups: (i) geometric inputs (S , AR , Λ_{LE}), nominally $\pm 2\text{--}5\%$ given that the published reference area excludes inboard fuselage panels; (ii) drag-build inputs ($C_{D_0} \in [0.015, 0.025]$, $e \in [0.85, 0.92]$ [14, 17]); and (iii) propulsion data ($\pm 3\%$ on quoted dry/AB thrust). Linearised sensitivity of the sustained turn rate to C_{D_0} at the level-turn condition ($V = 154.3$ m/s, AB thrust) gives $\partial\omega/\partial C_{D_0} \approx -68^\circ/\text{s}$ per unit C_{D_0} , so a ± 0.005 uncertainty in C_{D_0} alone yields $\pm 0.34^\circ/\text{s}$. Combining the three input

groups in quadrature gives a one-sigma band of approximately $\pm 2.5^\circ/\text{s}$ on the sustained turn rate and $\pm 25 \text{ m/s}$ on the rate of climb. The $\sim 5\text{--}21\%$ gap between the derived sustained turn rate and the lower end of cited values lies within this band; the $\sim 100\%$ gap in cited ROC does not.

VII. Avionics and Systems

The Su-57 incorporates the **Sh121 integrated radar complex**, comprising multiple active electronically scanned array (AESA) subsystems that together provide continuous hemispheric situational awareness [10, 21]. The array module counts and engagement capacities quoted below derive from Tikhomirov NIIP open-source disclosures aggregated by [10] and have not been independently confirmed:

- **N036-1-01 Byelka (Main Forward Array)** — An X-band AESA array with 1,514 transmit/receive (T/R) modules providing long-range target detection, tracking, and fire control [10]. Stated capability includes simultaneous tracking of up to 60 targets and engagement of up to 16 [10, 21].
- **N036B-1-01 (Side-Facing Arrays)** — Two lateral X-band AESA arrays, each with 404 T/R modules (2,322 T/R modules in the forward-hemisphere complex total), extending the radar field of regard to approximately $\pm 60^\circ$ beyond the forward sector [10].
- **N036L-1-01 (L-Band Wing Arrays)** — Leading-edge L-band AESA arrays providing counter-stealth detection capability, exploiting resonance effects at longer wavelengths ($\lambda \approx 15\text{--}30 \text{ cm}$) at which the radar cross section of low-observable aircraft — including the F-22 and F-35 — is significantly elevated compared to their X-band RCS [21, 22].

The entire Sh121 complex operates in Low Probability of Intercept (LPI) modes using frequency agility and low-power waveforms that reduce detectability by enemy radar warning receivers [21]. Sensor data from all radar arrays is fused with infrared search and track (IRST), electronic intelligence (ELINT), and electronic warfare (EW) inputs through the integrated avionics suite, providing the pilot with a unified tactical picture and reducing cognitive workload in contested environments [21].

A. Electronic Warfare Suite

The **L402 Himalayas** active jamming and electronic warfare system is housed in the tail cone and wing leading-edge structures [21]. The system provides:

- Passive detection and geolocation of emitters
- Active jamming of threat radars and datalinks
- Electronic support measures (ESM) for situational awareness

The **101KS Atoll** electro-optical system comprises the 101KS-V IRST with laser rangefinder for passive detection of aerial targets, and the 101KS-O Directed Infrared Countermeasures (DIRCM) system, which is believed to be the

first operational fitment of an active DIRCM on a fixed-wing fighter aircraft [21, 23]. The DIRCM blinds incoming IR-guided missiles with a focused infrared beam, providing a layer of active defence against all-aspect IR threats without expending expendable countermeasures.

B. Radar Cross Section

The Su-57 RCS reduction strategy is described in Section IV. The comparative table below places the Su-57 in context with peer aircraft [22]:

Table 2 Comparative RCS Values (X-band, Head-on, Clean Configuration) [22]

Aircraft	RCS (m^2)	RCS (dBsm)
Su-57 (Felon)	0.1–1.0	−10 to 0
F/A-18 Super Hornet	~ 1	~ 0
F-35 Lightning II	~ 0.005	~ -23
F-22 Raptor	$\sim 10^{-4}$	~ -40

Note: Values represent head-on X-band estimates for clean configurations with internal bays closed and are quoted as order-of-magnitude open-source consensus figures only; no measured cross-section data are publicly released for any of the platforms compared. Side and rear aspect RCS values are significantly higher across all platforms. L-band and VHF RCS values are higher for all aircraft, partially offsetting the X-band stealth advantage of the F-22 and F-35 [22].

VIII. Armament

The Su-57 features a comprehensive internal weapons architecture that preserves stealth characteristics across all mission profiles, eliminating the external drag penalties and RCS signature associated with external carriage [21, 23].

A. Weapons Bay Architecture

The aircraft incorporates four internal weapons bays [21, 23]:

- **Two Main Tandem Bays** — Primary bays arranged in tandem between the engines, each approximately $4.4 \text{ m} \times 0.9 \text{ m}$, housing up to four BVR missiles total.
- **Two Side Quick-Launch Bays** — Located at the wing roots, each housing one short-range air-to-air missile on a rapid-deploy launcher mechanism publicly demonstrated at Dubai Airshow 2025 [23].

Total internal hardpoints: 6. External hardpoints: 6 (stealth profile compromised when used) [21].

B. Internal Cannon

- **9A1-4071K GSh-30-1** — 30 mm autocannon embedded in the right LEVCON root, 150 rounds, rate of fire up to 1,500 rounds/min, weighing approximately 50 kg [21].

C. Air-to-Air Missiles

Table 3 Su-57 Air-to-Air Missile Loadout [21]

Missile	Guidance	Range	Role
R-77M	Active radar	Medium-BVR	Primary BVR, main bays
R-74M2	Infrared	Short-range	WVR, side bays, 75° off-boresight
R-37M	Active radar	300 km+	Very long-range BVR

D. Air-to-Ground and Anti-Radiation

Table 4 Su-57 Air-to-Ground Loadout [21]

Weapon	Range	Role
Kh-69 (Kh-59MK2)	~300 km	Precision strike, subsonic cruise missile
Kh-38M	Medium	Tactical air-to-ground, folding wings
Kh-58UShK	>200 km	Anti-radiation SEAD, Mach 3.6, folding fins
Kh-35U	Medium	Anti-ship
KAB-250L	—	Electro-optically guided bomb
KAB-500	—	Guided bomb

E. Defensive Systems

- **101KS-O DIRCM** — Directed infrared countermeasures, first operational fitment on a fighter aircraft [21, 23].
- **101KS-V IRST** — Passive infrared search and track with laser rangefinder [21].
- **L402 Himalayas** — Active electronic jamming suite [21].
- Chaff and flare dispensers.

F. Standard Combat Loadout

For a standard air-superiority mission with stealth preserved [21]:

- $4 \times$ R-77M (main tandem bays)
- $2 \times$ R-74M2 (side quick-launch bays)
- $1 \times$ 30 mm cannon (150 rounds)

IX. Strengths and Limitations

The assessments in this section combine the derived aerodynamic and performance results of Section VI with open-source platform data from Sections IV, V, and VII–VIII. Comparative statements against the F-22 and F-35 are

based on publicly reported figures for those platforms and inherit the same open-source uncertainty caveat as the Su-57 inputs.

A. Strengths

- **Supermaneuverability** — The LEVCON-TVC combination enables post-stall maneuvering at AOA exceeding 60° – 70° , supporting the Pugachev’s Cobra, Kulbit, and Herbst manoeuvres [13]. Malicki et al. [2] and Tsai et al. [3] confirm via CFD that LEVCON deflection provides 8.5–19.4% lift increment at high AOA, and Francis [12] demonstrates that integrated TVC sustains vehicle control through departure-prone regimes. This provides a decisive advantage in within-visual-range combat.
- **Counter-Stealth Capability** — The N036L L-band wing arrays exploit resonance at longer wavelengths to detect low-observable aircraft including the F-22 and F-35, partially offsetting the Su-57’s own stealth disadvantage in BVR engagements [21, 22].
- **Comprehensive Weapons Suite** — Internal carriage of diverse munitions across all mission profiles (BVR, WVR, SEAD, precision strike, anti-ship) with stealth preserved. The R-37M provides a 300 km+ BVR engagement capability [21].
- **DIRCM System** — The 101KS-O DIRCM, believed to be the first operational fitment on a fixed-wing fighter aircraft, provides active defence against all-aspect IR-guided missiles without expendable countermeasures [21, 23].
- **Supercruise** — Sustained Mach 1.3 cruise without afterburner is analytically confirmed in Section VI, with an excess thrust margin of 68.1 kN [15, 21].
- **Cost Position** — Open-source unit-cost estimates place the Su-57 below the F-35 and substantially below the F-22, although the absolute figures vary widely across sources and currency-conversion bases and are not reproduced here. The relative cost ordering is consistent across compilations and is the operationally relevant point [21, 26].

B. Limitations

- **Stealth Inferiority** — The Su-57’s head-on X-band RCS of 0.1 – 1.0 m^2 (Sukhoi patent disclosure, via [9]) is significantly higher than the F-22 ($< 0.0001\text{ m}^2$) and F-35 ($\sim 0.001\text{ m}^2$) [22]. The absence of serpentine inlet ducts is the primary contributor to this gap, as discussed in Section IV.
- **Engine Immaturity** — The AL-41F1 is an evolution of the Su-35’s powerplant rather than a clean-sheet fifth-generation design. The definitive Izdeliye 30, promising 176 kN and 20% improved specific fuel consumption, remained in development as of 2026 with no confirmed series production date [25].
- **Production Rate** — As of early 2026, approximately 30 airframes had been delivered against a 76-aircraft state contract, with few or no new aircraft identified as delivered in 2025 [24, 25]. This reflects industrial capacity constraints and sanctions impacts on avionics supply chains that cannot be resolved in the near term.

- **Sensor Fusion Gap** — While the Sh121 complex is technically capable, the Su-57’s sensor fusion and datalink architecture is reported to lag the F-35’s integrated battlespace picture, potentially increasing pilot workload in contested, multi-axis engagements [21].
- **Limited Combat Record** — Su-57 deployments in the Ukraine conflict have been confined to standoff engagements well beyond the forward line of troops, with no large-scale operational deployment reported [25]. This precludes independent assessment of real-world combat effectiveness and attrition robustness.
- **Export Failure** — India withdrew from the joint FGFA programme in 2018 citing unresolved performance and technology transfer concerns. Algeria remains the only reported prospective export customer, with delivery timelines uncertain [21, 26].

C. Overall Assessment

The Su-57 represents a genuine fifth-generation capability: world-class supermaneuverability quantified by independent aerodynamic analysis [2, 3, 13], a unique counter-stealth detection architecture in the N036L L-band arrays, and a comprehensive internal weapons suite supporting BVR, WVR, SEAD, and precision strike missions simultaneously. However, it embodies a design philosophy fundamentally different from Western counterparts — prioritising agility, versatility, and cost over the deep low-observable integration of the F-22 and F-35 [17, 22]. In a BVR engagement against Western fifth-generation fighters, the Su-57’s higher RCS places it at a detection disadvantage. In a WVR engagement, its LEVCON-TVC supermaneuverability architecture likely provides a decisive advantage [2, 13].

The program’s primary constraint is not aerodynamic or avionics capability, but industrial — the inability to produce sufficient airframes to translate genuine technical achievement into meaningful strategic air power mass [24, 25].

X. Conclusion

This paper has presented a comprehensive aerodynamic and systems analysis of the Sukhoi Su-57 (Felon) fifth-generation multirole fighter, employing USAF Digital DATCOM methodology [1] applied to publicly available geometric and performance data, supplemented by standard aerodynamic theory and recent peer-reviewed work on LEVCON aerodynamics [2–4].

A. Aerodynamic Findings

The following performance values were independently derived:

- **Lift Curve Slope:** $C_{L\alpha} = 2.728 \text{ rad}^{-1}$ (DATCOM method, half-chord sweep 38°) [1]
- **Subsonic Drag:** $D \approx 55.96 \text{ kN}$ at sea level, $V = 206 \text{ m/s}$, $C_D = 0.02732$ [16, 19]
- **Supercruise Excess Thrust:** $T_{\text{Excess}} \approx 68.1 \text{ kN}$ at Mach 1.3, confirming AL-41F1 capability for sustained supercruise [15]

- **Instantaneous Turn Rate:** $\dot{\psi} \approx 24.4^\circ/\text{s}$ at 206 m/s, $n = 9$ g, limited by pilot G-tolerance rather than aerodynamic limits [19, 20]
- **Sustained Turn Rate:** $\dot{\psi} \approx 23.7^\circ/\text{s}$ at 154.3 m/s (300 kts), maximum afterburner thrust with LEVCON and 3D-TVC augmentation [2, 12, 20]
- **Rate of Climb:** $ROC \approx 180.5$ m/s (34,780 ft/min) at TWR = 1.09, sea level [19]

B. Performance Discrepancy Observation

A consistent discrepancy was identified between derived performance values and figures widely cited in open literature:

Table 5 Derived vs. Cited Performance Figures [19–21]

Parameter	Derived	Cited	Discrepancy
Sustained Turn Rate	$23.7^\circ/\text{s}$	$25\text{--}30^\circ/\text{s}$	$\sim 5\text{--}21\%$
Rate of Climb	180.5 m/s	350–384 m/s	$\sim 94\text{--}113\%$

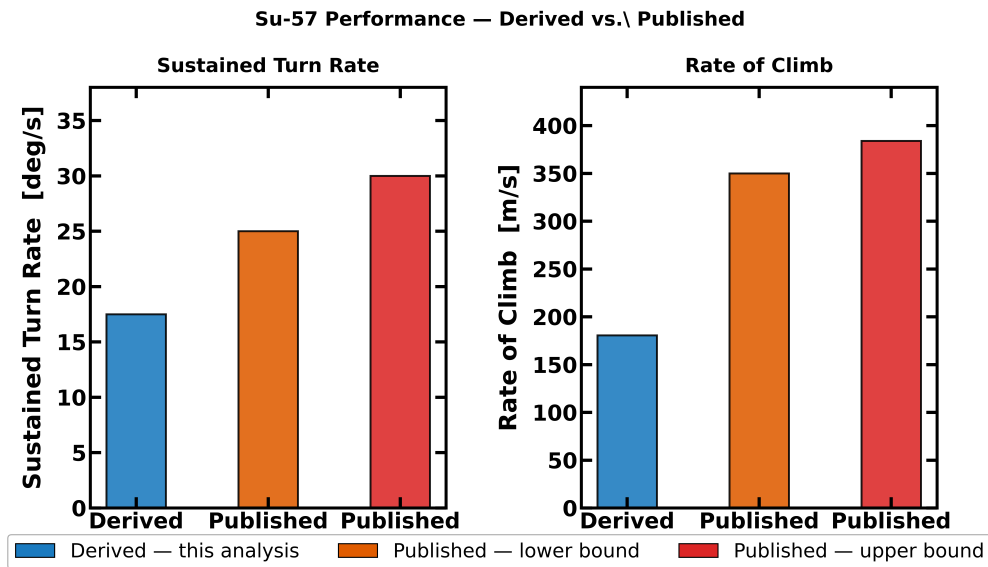


Fig. 12 Derived vs. published performance — sustained turn rate and rate of climb [19, 21].

Figure 12 compares the derived Su-57 performance estimates with published open-source reference values for sustained turn rate and rate of climb.

The published component specifications — 142.2 kN afterburning thrust per engine and TWR = 1.09 — are internally consistent. The derived sustained turn rate of $23.7^\circ/\text{s}$ at 300 kts using maximum afterburner thrust falls within $\sim 5\text{--}21\%$ of the lower end of the cited $25\text{--}30^\circ/\text{s}$ range, well within the combined uncertainty on C_{D_0} , e , and the LEVCON lift increment, and does not require invoking unpublished thrust capability.

A more substantial discrepancy persists for rate of climb. The cited ROC of 350–384 m/s would require approximately 506–549 kN total thrust, approximately double published figures, which is physically implausible with any known AL-41F1 variant [19, 21]. The most plausible explanations are that the cited figures correspond to unspecified flight conditions (light weight, partial fuel, optimum altitude zoom-climb) or originate from unverified secondary sources.

C. Overall Assessment

The Su-57 represents a genuine aerodynamic achievement. Its LEVCON-TVC supermaneuverability architecture is unique among operational fifth-generation fighters and has been independently validated by multiple peer-reviewed computational and experimental studies [2–4, 13]. Its N036L counter-stealth L-band arrays represent an innovative asymmetric response to Western low-observable designs [21, 22]. The comprehensive internal weapons suite supports the full spectrum of air superiority, SEAD, and precision strike missions while preserving stealth.

The aircraft’s primary operational constraint is not technical capability but industrial — production rates remain insufficient to translate genuine aerodynamic and systems excellence into strategic air power mass [24, 25].

All values in this paper are derived from publicly available data using USAF Digital DATCOM methodology [1] and standard aerodynamic equations. They represent engineering approximations and should not be treated as definitive specifications.

D. Known Limitations and Future Work

- **LEVCON Geometric Data** — The LEVCON load-factor increment ($\delta n_{\text{LEVCON}} \approx +0.82$, Eq. ??) was estimated by applying the lower-bound +15% C_L augmentation reported by Malicki et al. [2] to the base load factor. A first-principles derivation requires detailed LEVCON planform geometry not available in the public domain.
- **Transonic Drag Rise Not Modelled** — $C_{D_{\text{wave}}}$ at supercruise was treated as a constant (0.02) rather than derived from a Mach-dependent transonic drag rise model. A Kármán–Ogive or Lock approximation [16] accounting for the drag rise between M_{crit} and Mach 1.3 will be incorporated in a future revision.
- **Thrust Lapse Model Absent** — All propulsion calculations assume sea-level static thrust scaled by published TWR. An altitude-dependent thrust lapse model of the form $T = T_0 \cdot (\rho/\rho_0)^n$ will be incorporated in a future revision [17].
- **AVL/OpenVSP Vortex-Lattice Cross-Validation** — DATCOM methods are known to be conservative for blended wing-body configurations with LEVCON augmentation [1]. A vortex-lattice validation using AVL or OpenVSP is planned as a higher-fidelity cross-check, as suggested by Flyward_Aerospace (r/aerospace, March 2026).
- **Izdeliye 30 Engine Not Modelled** — All propulsion calculations use AL-41F1 specifications. A performance comparison with the Izdeliye 30 will be included when verified public data become available [25].

- **Control Surface Derivatives from Open Literature** — $C_{l_{\delta a}}$, $C_{n_{\delta r}}$, C_{l_p} , and C_{n_r} are retained from open literature as first-principles derivation requires detailed control surface geometry unavailable in the public domain [1, 18].

Acknowledgments

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